

# NICHOLAS FOLEY

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## Summary

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Marine Corps veteran, former USAA intern, and Computer Science PhD student who builds and validates machine learning and LLM systems under real-world latency, cost, and reliability constraints. Published researcher and co-owner of an AI consulting firm who documents methods for technical review and explains results plainly to non-technical decision-makers.

## Experience

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### Foresyth Foley LLC

June 2026 – Present

*Co-Owner & Machine Learning Consultant*

*San Antonio, TX*

- Scoped the migration of T9, a telecom client's production competitive-intelligence system, from a hosted commercial LLM API to a self-hosted open-weight model; wrote the statement of work; currently evaluating candidate models against the client's output-quality, latency, cost, and data-privacy requirements.

### VAIL Lab, UT San Antonio

June 2025 – Present

*Graduate Research Assistant*

*San Antonio, TX*

- Designed, trained, and validated QSV, a quaternion attention architecture for vision on spherical lattices, benchmarked against established baselines under matched conditions; first author of the resulting paper, currently under peer review.
- Write clean, documented Python for experiment pipelines; log every run's hyperparameters, metrics, and artifacts in Weights & Biases so results are reproducible and reviewable.

### CloudSys Lab, UT San Antonio

January 2025 – June 2026

*Graduate Research Assistant (SDS Research Fellow, Jan–Aug 2025)*

*San Antonio, TX*

- Benchmarked neural-network partitioning strategies on a 6-device NVIDIA Jetson and Raspberry Pi cluster and mapped the accuracy–latency–power Pareto front, separating deployable setups from dominated ones (IEEE EdgeCom 2025).
- Built Python data pipelines and trained a graph neural network to predict the best partitioning strategy.

### USAA

May 2024 – August 2024

*Software/DevOps Engineer Intern*

*San Antonio, TX*

- Automated Aurora PostgreSQL grant and permission provisioning with SQL scripts, replacing a manual deployment step.
- Adapted Java deployment tooling for a cloud migration within USAA's SDLC and change-management controls.

### United States Marine Corps, VMGR-352

September 2015 – September 2020

*Aircraft Collateral Duty Quality Assurance Representative*

*San Diego, CA*

- Performed risk-based inspections for regulatory compliance, translated findings into recommendations for non-technical leadership, and designed and taught quality-standards courses to 100+ Marines.

## Selected Projects

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### Research Knowledge Navigator ([github.com/Foley-ops/knowledge-navigator](https://github.com/Foley-ops/knowledge-navigator))

2026

- Built a containerized knowledge-graph wiki of math and AI concepts with retrieval-augmented Q&A: a local LLM answers from up to 8 retrieved pages with citable IDs, and any answer with invalid JSON or citations is discarded.
- Added an evaluation harness to choose the answering model from 8B–12B open-weight candidates; search, comparison, and path-finding stay deterministic and never call the model.

### Trading-Card Recognition System (Magic: The Gathering)

2026

- Built a local card-identification pipeline (detect, warp, embed with DINOv3, FAISS nearest-neighbor search), replacing a perceptual-hash baseline whose match distributions overlapped too much to scale.
- Evaluated on a 5,327-card catalog with 2,060 distorted queries (Recall@1 0.82, Recall@5 1.00), traced remaining errors to reprints, and found ViT beats ConvNeXt on shared-frame cards because average pooling washes out the art.

## Education

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University of Texas at San Antonio — *PhD, Computer Science* | GPA: 4.0

August 2025 – Present

University of Texas at San Antonio — *BS, Computer Science* | GPA: 4.0 | President's List 2023–2025

May 2025

## Publications & Manuscripts

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- *Where to Split? A Pareto-Front Analysis of DNN Partitioning for Edge Inference.* Masud, **Foley**, et al. IEEE EdgeCom 2025.
- *Transport, Not Routing: Dissecting Quaternion Attention for Vision.* **Foley** (first author). Under review.

## Skills

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- **Classical ML:** linear/logistic regression, SVMs, decision trees, random forests, k-means, hierarchical clustering, DBSCAN
- **Deep Learning & LLMs:** PyTorch, GNNs, Transformers; RAG, output validation, prompt engineering, Ollama, Hermes, MCP
- **Statistics:** EDA, statistical inference, hypothesis testing, confidence intervals, experimental design
- **Programming & Data:** Python (since 2018; scikit-learn, XGBoost, pandas, NumPy), SQL (PostgreSQL, SQL Server), C/C++
- **MLOps & Cloud:** Docker, Docker Compose, Git, CI/CD, Weights & Biases, MLflow, AWS (Aurora, EC2), GCP, Azure, Linux